

Ilaria

An open Romanian-first language model with one-shot memory — powering Swypik

THERAPIUM GROUP SRL · Romania · founded 2025

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Code: github.com/office233/ilaria · Product: swypik.com

Problem

- Romanian is a minor share of the training data of today's open models (Llama, Qwen, Gemma).
- Frozen LLMs cannot learn a new fact without retraining — but products, prices and policies change daily.
- Romanian and CEE SMEs lack affordable AI that speaks their language and stays up to date.

Solution: Ilaria

- Bilingual Romanian–English decoder trained from scratch (RoPE, SwiGLU, 32k byte-level BPE).
- Own Go inference engine, numerically identical to PyTorch (max $|\Delta \text{logit}|$ $2e-5$).
- Hippocampus-inspired episodic memory: learns a fact from ONE exposure, no gradient update,
- and keeps it after restart.

Traction so far (measured)

- Ilaria-130M trained on 3.0B tokens (FineWeb-2 Romanian, FineWeb-Edu, Wikipedia) in ~4 h on one GPU.
- Held-out perplexity on 2025 Wikipedia: 24.4 overall — Romanian 20.8, English 28.7.
- One-shot continual-learning benchmark: 89% strict accuracy vs 1% for the same frozen model;
- 100% recall after restart. Open source (AGPL-3.0).

Product: Swypik

- Video-native social commerce for Romania and Central-Eastern Europe (swypik.com), plus a Romanian ERP.
- Ilaria's jobs inside Swypik: content moderation, Romanian shopping assistant,
- captions and translation, product-quality scoring.
- One-shot memory = new products and prices are known instantly, without retraining.

Market

- ~24M Romanian speakers, plus CEE e-commerce and SMEs needing local-language AI.
- Beachhead: our own platform (Swypik) → then Romanian SMEs via API and the ERP.
- Open models + low-cost inference (small model, Go engine) keep serving costs low.

Why we win

- Romanian-first data and tokenizer (1.67 tokens/word in Romanian vs 3.11 for our previous tokenizer).
- Continual learning without retraining — a capability frozen LLMs structurally lack.
- Full stack in-house: data pipeline, trainer, inference engine, memory, product.

NVIDIA technology

- Training: PyTorch + CUDA (bf16, torch.compile) on NVIDIA data-centre GPUs.
- Inference: our Go engine calls cuBLAS on NVIDIA GPUs.
- Next: EuroHPC application for H100 time on MareNostrum 5 (1B → 3B → 7B);
- serving with TensorRT-LLM / NIM.

Roadmap

- Q4 2026: Ilaria-1B and 3B (gated scaling ladder), instruction tuning for Romanian assistant tasks.
- Q1 2027: Ilaria-7B on up to 300B tokens; open release of weights, tokenizer, benchmark.
- 2027: Ilaria in production across Swypik; API for Romanian SMEs.

Team & ask

- Abel Varga — Founder & CEO, builds the full stack (models, engine, platform).
- Oana Cioacata — CFO.
- Ask: GPU compute and cloud credits to train and serve Ilaria at scale;
- NVIDIA Inception support for training, inference optimisation and go-to-market.